P-P-105C
September 28, 1970
SUPERSEDING
Fed. Spec. P-P-105B
September 10, 1964

FEDERAL SPECIFICATION

PAPER, ABRASIVE (FLINT)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. Flint paper covered by this specification is a paper coated on one surface with a flint abrasive.

1.2 Classification.

1.2.1 Classes and grit sizes. Flint paper shall be of the following classes and grit sizes, as specified (see 6.2):

Class 1 - Sheets, flint finishing paper (see 3.4.1).
  Grit size - 320, 280, 240, 220, 180, 150.
Class 2 - Sheets, flint paper (see 3.4.1).
  Grit size - Extra fine, fine, medium, coarse, extra coarse.
Class 3 - Rolls, flint paper (see 3.4.2).
  Grit size - Extra fine, fine, medium, coarse.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-P-31 - Paper, General Specifications and Methods of Testing.

Federal Standards:

Fed. Std. No. 102 - Preservation, Packaging and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents. U.S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.)

Military Specifications:

MIL-P-3816 - Packaging and Packing of Abrasives and Abrasive Products for Shipment and Storage.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposed shall apply.

Department of Commerce Commercial Standard:

PS8 - Grading of Abrasive Grain on Coated Abrasive Products.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Abrasive.

3.1.1.1 Flint. The abrasive shall be crushed or ground from clean hard quartz, quartzite, or flint and shall be relatively free from extraneous materials.

3.1.1.2 Grading, control, and tolerance. Grading of abrasive particles shall be in accordance with PS8; and the tolerance therein shall apply to the abrasive grain recovered from the coated abrasive product.

3.1.2 Backing. The backing shall be paper and shall meet the requirements of tables I and II directly after being conditioned at 70° ± 2°F. and 50 ± 2 percent relative humidity for at least 24 hours.

TABLE I. Grit sizes and weight of paper backing

Class No.	Grit size	Nominal paper weight, 24 by 36 inches, 480 sheets	Tolerance	Symbol or designation
		Pounds	Pounds	
1	320 through 150	40	37.5 - 42.5	A
2	Extra fine, fine	50 to 60	46.5 - 63.0	-
2	Medium, coarse, extra coarse	55 to 60	51.0 - 63.0	-
3	Extra fine, through coarse	130	122.0 - 135.0	E

3.1.3 Adhesive. The adhesive shall be a good quality, animal hide glue base, suitable for the purpose intended, and shall conform to MMM-A-100, adhesive, animal glue.

3.2 Coating.

3.2.1 Class 1. The flint finishing paper sheets shall be open coat. The abrasive flint grain shall be applied to one side of the paper backing only, and cover approximately 50 percent of the backing, thus leaving open spaces between the grains of abrasive (see 6.4).

3.2.2 Classes 2 and 3. The flint paper sheets and rolls shall be closed coat. The abrasive flint grain shall be applied to one side of the paper backing only, and shall completely cover the surface of the backing (see 6.4).

3.3 Tensile breaking strength. The average breaking strength of coated abrasive paper shall be in accordance with table II, when tested as described in 4.4.2.

TABLE II. Tensile breaking strengths

Class No.	Grit size	Tensile strength, average	
		Machine direction	Across machine direction
		Pounds/Inch	Pounds/Inch
1	320 through 150	20	10
2	Extra fine, fine	35	15
	Medium, coarse, extra coarse	40	20
3	Extra fine through coarse	85	45

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3.4 Form and dimensions.

3.4.1 Sheet. Unless otherwise specified (see 6.2), flint paper shall be furnished in the following sizes, with a tolerance of ± 1/8 inch in length and ± 1/16 inch in width, when measured at, and after conditioning at 70° ± 2°F. and 50 ± 2 percent relative humidity for at least 24 hours:

Class 1 - 9 by 11 inches
Class 2 - 9 by 10 inches

3.4.2 Roll. Class 3 flint paper shall be furnished in rolls with a standard length of 50 yards with a tolerance of minus "0" plus 12 inches and in the widths as specified (see 6.2).

3.5 Identification marking. Each sheet of class 1 and 2 abrasive paper shall be marked with the manufacturer's name or trade mark (of such character that the source may be readily determined), and the grit size. Each roll of class 3 abrasive paper shall be marked with the manufacturer's name or trade mark, and the grit size at least once on every yard on rolls 1 inch and wider. All markings shall be legible and permanent.

3.6 Workmanship. Abrasive flint paper shall be free from imperfections affecting its appearance or which may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. The sample size levels of inspection shall be in accordance with MIL-STD-105.

4.2.1 Lot. All abrasive paper of the same class and grit size manufactured under essentially the same conditions and offered for delivery at one time shall be considered a lot for the purposes of examinations and tests.

4.2.2 Sampling for examination of packaging and packing. Packaged and packed abrasive paper in shipping containers shall be sampled in accordance with inspection level I.

4.2.3 Sampling for visual, dimensional, and workmanship characteristics of end item. Sampling for these characteristics shall be in accordance with inspection level S-4.

4.2.4 Sampling for tests. Samples for test shall be selected in accordance with inspection level S-2. Samples may be selected from those selected in accordance with 4.2.3.

4.3 Examination and tests. The acceptable quality levels (AQL's) shall be in accordance with MIL-STD-105.

4.3.1 Examination.

4.3.1.1 Packaged and packed abrasive paper. Abrasive paper, ready for shipment, selected in accordance with 4.2.2, shall be examined with AQL's and classification of defects in accordance with MIL-P-3816.

4.3.1.2 Individual sheets and rolls. Samples of abrasive paper selected in accordance with 4.2.3 shall be examined in accordance with table III. The AQL's for defects listed in table III shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects. The base weight of the paper shall be as specified in 3.1.2.

TABLE III. Classification of defects, end item

Examine (applicable paragraph)	Defects	Major	Minor
1.2.1	Size of grit not as specified or ordered	X	
3.1.1.1	Extraneous material present in abrasive	X	
3.2	Abrasive not closed or open coat as specified	X	
3.2	Coating not completely covering surface of backing	X	
3.4.1	Dimensions of sheets, not correct size or not within tolerances	X	

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TABLE III. Classification of defects, end item (con.)

Examine (applicable paragraph)	Defects	Major	Minor
3.4.2	Size of rolls not as specified or ordered	X	
3.5	Identification marking, ommitted, illegible, incomplete, or incorrect		X
3.6	Sheets and rolls, torn or cracked	X	

4.3.2 Lot acceptance tests. Samples selected in accordance with 4.2.4 shall be subjected to the tests in 4.4, AQL 10.0 percent defective.

4.4 Tests.

4.4.1 Grain size.

4.4.1.1 Grading for grain size. Grading to determine the grain size, including recovery of abrasive grain from the sheets, preparation of grain for grading, and procedures shall be in accordance with PS8.

4.4.1.2 Certification of Grading. Grading tests specified in 4.4.1.1 shall be omitted in instances in writing that the abrasive grain complies with the requiremsnts of PS8 for flint. Contractors are urged to include this certification with material offered for inspection and inspectors are urged to determine whether such certification is included with such material in order to preclude the necessity for grading tests.

4.4.2 Breaking strength. The tensile breaking strength of the coated abrasive, directly after conditioning for at least 24 hours at 70° ± 2°F. and 50 ± 2 percent relative humidity shall be determined by method 171 of UU-P-31, except that the jaws of the testing machine shall be 5 inches apart, and that six specimens 1 inch wide shall be taken both longitudinally and transversely from sample sheets and rolls. The average tensile strength in both directions shall be computed from at least five specimens. Samples tearing at the jaws of the tensile testing machine should be discarded (see 4.3).

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing of abrasive paper shall be in accordance with MIL-P-3816, as specified for level A, B, or C (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special markings required by the contract or order, packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. In addition to any special markings required in the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Abrasive flint paper is intended for all ordinary hand and machine sanding of wood or painted surfaces.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class and grit size for sheets (see 1.2).
(c) Class, grit size, and width of rolls (see 1.2 and 3.4.2).
(d) Level of packaging and level of packing required (see 5.1).

6.3 Flint. A white, translucent, somewhat crystalline, variety of natural silicon dioxide, having a structure such that when crushed develops sharp edges. The hardness is approximately 6.8 - 7.0 (Mohs scale), or 820 Knoop scale, with a specific gravity of approximately 2.6.

6.4 Coating. A closed coat is one in which the abrasive grain is put on in such quantity as entirely to cover the backing. An open coat covers about 50 percent of the backing thus leaving open spaces between the particles of abrasive grain.

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6.5 Work value. Coated abrasives should be stored in a cool even temperature area where the relative humidity is somewhere between 40 to 50 percent. Do not store on concrete floors. The best work value will be obtained if stored and worked in rooms having a relative humidity of approximately 50 percent.

6.6 Abrasive grain size. The size of abrasive grain is indicated in table IV which shows the approximate relation in size between flint garnet, and artificial abrasives.

6.6.1 Cross reference for grit sizes. Product Standard PS8, Grading of Abrasive Grain on Coated Abrasive Products uses the same classification for flint grit, silicon carbide, and aluminum oxide abrasives. Table IV shows the comparative grit sizes. Commercial producers have also adopted this classification.

TABLE IV. Comparative grit sizes

Silicon carbide. aluminum oxide, garnet and flint grit (class 1) sizes.	Classes 2 and 3 flint
320	
280	
240	
220	Extra fine (EF)
180	
150	Fine (F)
120	
100	
80	Medium (M)
60	
50	Coarse (C)
40	
36	Extra coarse (EC)
30	

CUSTODIANS:

Army - MR
Navy - YD
Air Force - 84

Review activity:

Navy - SH

User activities:

Navy - MC
Army - AV

Preparing activity:

GSA-FSS

Military Coordinating Activity:

Navy - YD

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, and mail to General Services Administration., FSS (FMSH), Washington, D.C. 20406

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

P-P-105C Paper, Abrasive (Flint)

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

GSA DC 69-9761

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